



Bachelor of Science in Informatics and Computer Science

ICS 4111: EMBEDDED SYSTEMS AND INTERNET OF THINGS

CAT 2

The link layer of the IoT stack has several protocols that define aspects such as the network type, cryptography technique, range of coverage, power consumption and data throughput of the IoT architecture. Such protocols include WLAN (Wi-Fi), Z-Wave, Bluetooth Low Energy, Zigbee, GSM (2G), GPRS (2.5G), UMTS (3G), LTE, 4G, 5G, Narrowband-IoT, Sigfox, NFC, LoRa & MLoTy. It is crucial for IoT developers to compare these protocols to determine the most appropriate communication technology when developing a system.

Instructions:

1. **In your defined groups for the semester project**, prepare a 10 to 15-minute presentation on the topic as assigned by the instructor.
2. The presentation should include a comparison of the technical specifications of the communication protocols assigned. Sample specifications for comparison have been provided in the reference research papers on e-learning.
3. As part of the semester project deliverable, you are expected to prepare power budgets of the IoT based greenhouse monitoring system. Using the components specified in deliverable 1, create two sample architectures that run on the two protocols assigned to your group. Thereafter, develop power budgets of the embedded devices running on these architectures and suggest measures for suitable power consumption. A walkthrough of how to prepare a power budget has been provided on e-learning.
4. Include an element of creativity in the presentation: use of colours, interesting fonts, appealing and appropriate images and videos etc. are highly encouraged and will be assessed during the presentation.

Once done, you will **present your work to the class** at your scheduled time as listed. **All members are expected to be present during presentation.** There will be a score for teamwork which will be assessed during the presentation. **For the groups that will be presenting during the online classes, speakers are expected to have their cameras on during their presentation.**

After presenting in class, submit your slide deck to e-learning. Name your file to *ClassGroup_Project Group Name_Name of Topic* e.g. *4A_Our Cool Group_LoRavsWiFi* and submit as a PDF document.

Feel free to use any available resources for your work, however the rules against plagiarism apply. **Do not directly use ChatGPT or any other AI-generative text applications to develop your work (especially in generating the slides)**, this will be penalised strictly when detected.